

ITERATION 3

Objective

Extend your **Level 1 working game (Iteration 2)** into a **complete, polished game system** with:

- Multiple levels (progressive difficulty)
- Bonus/power-up system
- Complete game flow (start → play → end → restart)
- Enhanced user experience

This is where your project becomes a “**real game**”, not just a working prototype.

SCOPE OF ITERATION 3

You must implement:

- **3 Levels total**
 - Increasing difficulty per level
 - Bonus item system
 - Complete game lifecycle:
 - Win progression
 - Game over
 - Replay / return to menu
-

DETAILED MARKING BREAKDOWN (30 MARKS)

1. MULTI-LEVEL SYSTEM (10 Marks)

Requirements

- Level 1 (reuse from Iteration 2)
- Add **Level 2 and Level 3**
- Each level must feel **different and harder**

Marking

Component	Marks
Level 2 implemented	2
Level 3 implemented	2
Speed increase per level (ball)	2
Different brick layouts per level	2
Smooth transition between levels	2

2. POWER-UP SYSTEM (10 Marks)

Requirements

- Bonus appears **randomly when a brick breaks**
- Falls downward toward paddle
- Paddle can collect it

Minimum: 3 bonus types required

Examples

- Slow Ball
- Fast Ball
- Extra Life
- Paddle Size Increase

Marking

Component	Marks
Random spawn logic	2
Falling animation	2
Paddle collision detection	2
At least 3 working bonus types	4

Static or fake bonus (no effect) → marks deducted

3. COMPLETE GAME FLOW (5 Marks)

Requirements

- Player progresses:
 - Level 1 → Level 2 → Level 3
- After final level:

- Show **Win Screen**
- If lives end:
 - Show **Game Over Screen**

Marking

Component	Marks
Level progression works	2
Game Over screen	1
Win/Completion screen	1
Return to menu / replay option	1

4. GAME POLISH & STABILITY (5 Marks)

Requirements

- Smooth gameplay experience
- No flickering/glitches
- Balanced difficulty

Marking

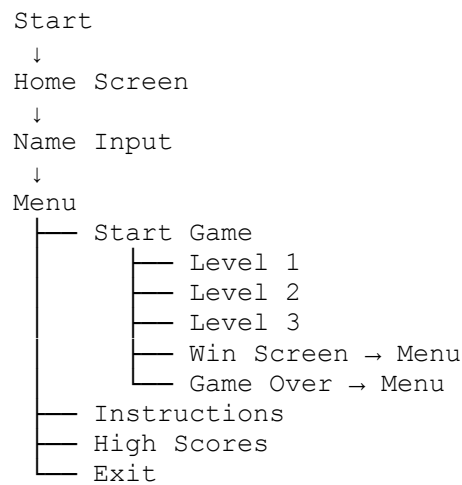
Component	Marks
No crashes/freezes	2
Smooth rendering	1
Balanced gameplay	1
Clean transitions	1

TECHNICAL EXPECTATIONS

Students must now demonstrate:

- Reusable procedures (Level handling, bonus logic)
- Proper state control:
 - Current level
 - Active bonuses
- Efficient updates (no full redraw flickering if avoidable)

EXTENDED GAME FLOW (FINAL ARCHITECTURE)



BONUS MARKS (+10)

Students can earn additional marks for advanced features:

Feature	Marks
File handling (save/load scores)	+3
Pause system (key press to pause/resume)	+2
Sound effects (PC speaker)	+2
Mouse support	+2
Visual effects (animations/trails)	+1

SUBMISSION REQUIREMENTS

1. Code

- Fully working **3-level game**

2. Documentation

Must include:

- Level design explanation
- Bonus system logic
- Game flow explanation

3. Video Demo (5–8 minutes)

Must show:

- Level progression
- Bonus usage
- Game over scenario
- Winning the game